

# 5. Managing the Test Activities

## A. Test planning

Release planning: backlog, testable stories, test approach

Iteration 1

Iteration 2

Iteration 3

Iteration 4

Iteration planning: story risks, testability, tasks, estimates

A **test plan** describes the test objectives, resources, processes and schedule, and is a way to communicate them. It typically covers the context (scope, objectives, constraints, test basis), assumptions, stakeholders, communication, the risk register, the test approach, and the budget and schedule.

In iterative development, testers help plan releases and iterations: they write testable stories and acceptance criteria, take part in risk analysis, estimate test effort and plan the test approach.

## B. Test case prioritisation

Test cases are put into an order for execution. Common approaches:

- **Risk-based:** tests for the most important risks run first
- **Coverage-based:** tests that achieve the most coverage (or the most **additional** coverage) run first
- **Requirements-based:** tests for the highest-priority requirements run first

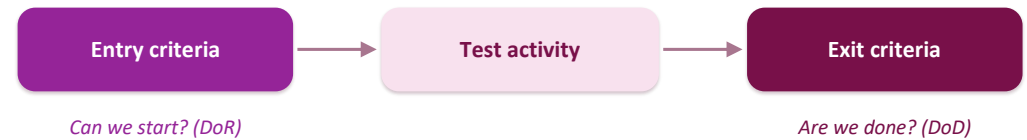
**Exam tip:** Dependencies come first: if a high-priority test depends on a lower-priority one, run the lower-priority test first.

## C. Testing quadrants



The quadrants group tests by whether they are **business-facing** or **technology-facing**, and whether they **support the team** (guide development) or **critique the product** (check it against expectations). They help make sure all the right test types are covered.

## D. Entry and exit criteria



**Entry criteria** say when an activity can start; **exit criteria** say when it is finished. In Agile they are often the Definition of Ready and the Definition of Done.

- **Entry examples:** resources and testware are available; the test object is of good enough initial quality (e.g., smoke tests pass)
- **Exit examples:** the required coverage is reached; defect counts and density are acceptable; planned tests have run; defects are reported; regression tests are automated

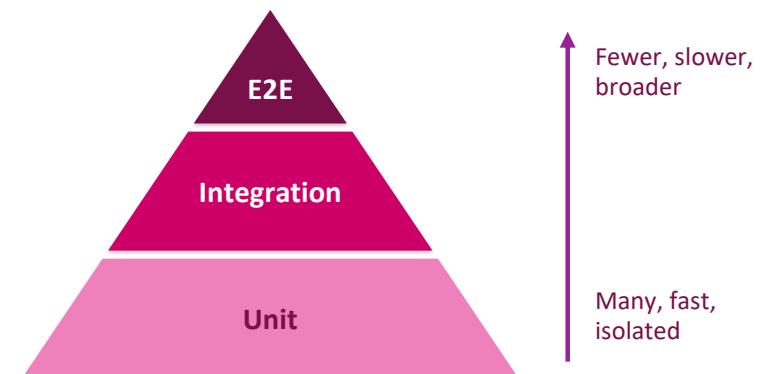
Running out of time or budget can also be a valid reason to stop, if stakeholders accept the risk.

## E. Estimation techniques

- **Ratios:** use ratios from past projects, e.g., the ratio of development effort to test effort
- **Extrapolation:** measure early in the project, then extrapolate, e.g., using the average of past iterations
- **Wideband Delphi:** experts estimate separately, discuss the differences, then re-estimate until they agree (**Planning Poker** is a variant)
- **Three-point estimation:** from optimistic (a), most likely (m) and pessimistic (b) estimates,  $E = (a + 4m + b) / 6$  and  $SD = (b - a) / 6$

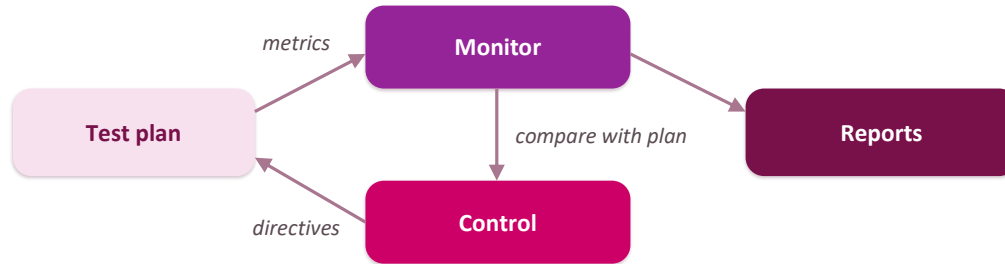
Every estimate rests on assumptions. Break big tasks into smaller ones to estimate more accurately.

## F. Test pyramid



The test pyramid shows tests at different levels of detail: **many** small, fast, isolated tests at the bottom and **few** slow end-to-end tests at the top. It helps decide where to automate and how to share out the effort.

## G. Monitoring, control and metrics



**Test monitoring** gathers information to compare progress with the plan. **Test control** uses it to take corrective action. **Test completion** collects the results at the end of a stage.

**Common metrics:** project progress, test progress, product quality, defects, risk, coverage and cost.

## H. Configuration management



**Configuration management** keeps track of test work products such as plans, test cases, scripts, results and environments. Each item is uniquely identified, version-controlled and linked to related items. Once approved, an item becomes a **baseline** that can only be changed through formal change control.

## I. Defect management and reports



Each reported anomaly follows a workflow: it is logged, analysed and classified, a response is decided (fix, defer or reject), and it is closed. Not every anomaly is a real defect; some are false positives or change requests.

**A defect report typically includes:** an ID and title; date and author; the test object and environment; context; steps to reproduce, with evidence; expected and actual results; **severity** (how much impact); **priority** (how urgently to fix); status; and references.

## J. Risk and risk level

Likelihood ↑

|      |        |           |           |
|------|--------|-----------|-----------|
| High | High   | Very high | Very high |
| Med  | Medium | High      | Very high |
| Low  | Low    | Medium    | High      |
|      | Low    | Med       | High      |

Impact →

A **risk** is something that might happen and would have a negative effect. Its **risk level** comes from its **likelihood** and its **impact**.

**Risk analysis** = identifying and assessing risks.

**Risk control** = mitigating and monitoring them.

## K. Project vs product risks

**Project risks** affect the project's ability to deliver:

- **Organisational:** late deliveries, inaccurate estimates, cost-cutting
- **People:** missing skills, conflict, communication problems, staff shortages
- **Technical:** scope creep, poor tool support
- **Supplier:** a third party fails to deliver

**Product risks** affect the quality of the product:

missing or wrong functionality, calculation or runtime errors, poor performance, a poor user experience, or security weaknesses. Possible impact: unhappy users, lost revenue and reputation, even harm to people.

## L. Product risk analysis and control

Product risk analysis shapes the testing: its **scope**, the test levels and types, techniques, coverage, effort and the order of tests.

**Ways to respond to a risk:** mitigate it by testing, accept it, transfer it, or have a contingency plan.

**Mitigating risk through testing:** testers with the right skills, the right level of independence, reviews and static analysis, suitable techniques and coverage, the right test types, and dynamic and regression testing.

## M. Test reports and communication

- **Test progress report** (regular, during testing): progress and deviations from the plan, impediments, metrics, new or changed risks, and plans for the next period
- **Test completion report** (at the end of a stage): a summary, an evaluation of quality, deviations from the plan, impediments, metrics, unresolved risks and defects, and lessons learned

Choose how to communicate for the audience: conversations, dashboards, email or chat, online documents, or formal reports.